

• SYSTEM ARCHITECTURE • DIAGRAM-FIRST

System architecture.

Intent-firewall copilot for agentic Sui DeFi. A keyless agent builds one unsigned PTB, a deterministic Guardian re-derives the math and fail-closes, and your wallet signs the literal artifact you reviewed.

14

GUARDIAN GATES

0

USER-FUND KEYS SERVER-SIDE

3

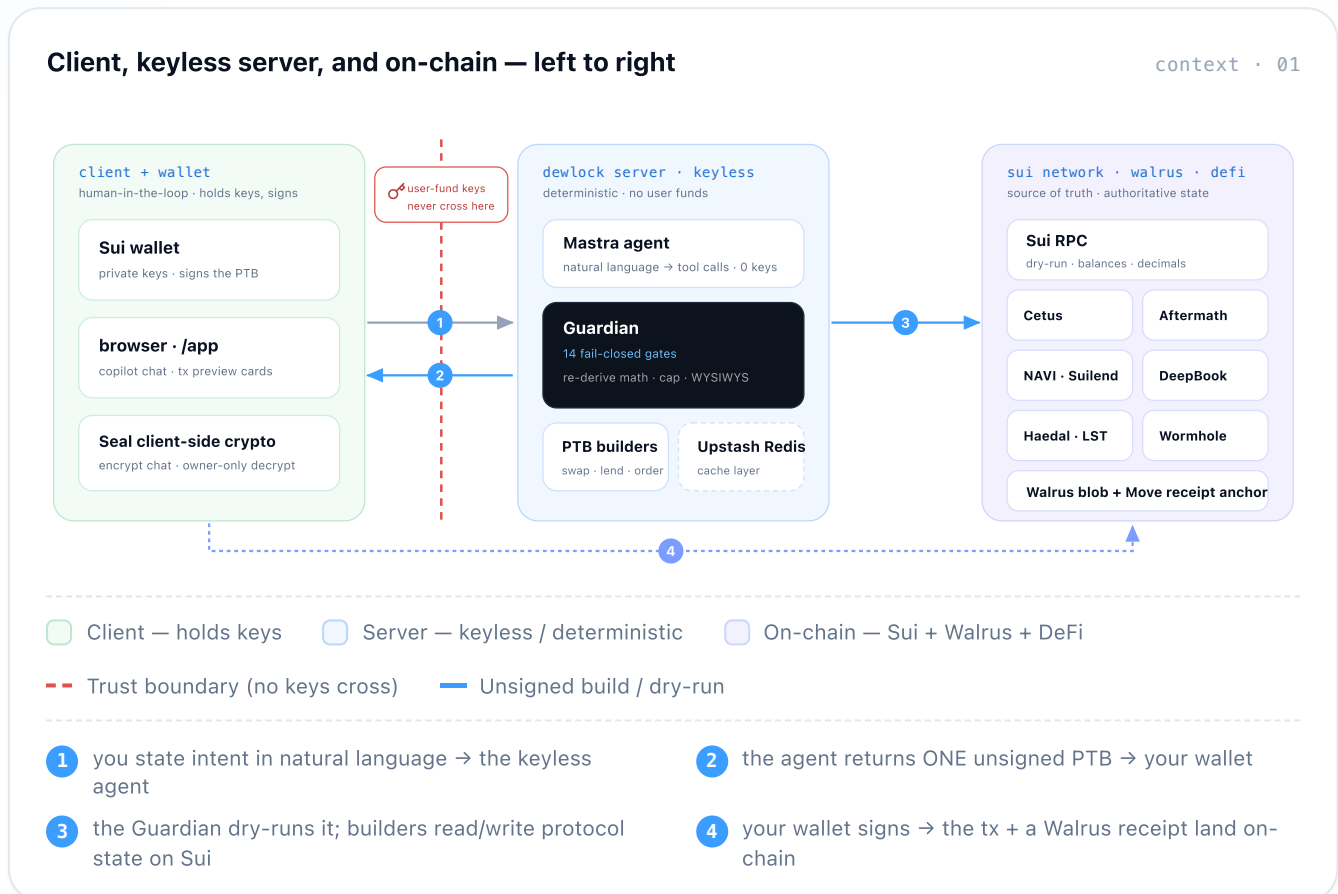
TRUST ZONES, ONE BOUNDARY

1137

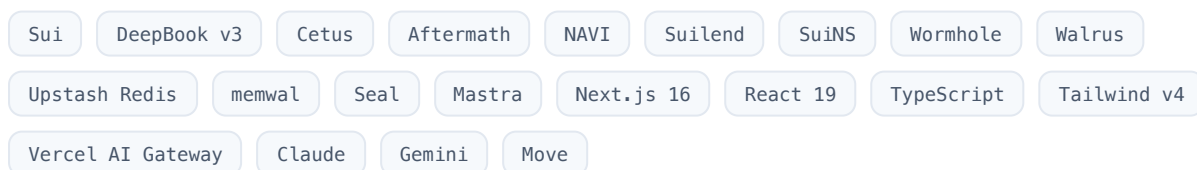
TESTS GREEN

Three trust zones, one boundary keys never cross.

The agent and Guardian run server-side but stay **keyless** — they only ever emit an *unsigned* bundle. Signing happens in your wallet; settlement, valuation and receipts happen on Sui and Walrus.



• STACK

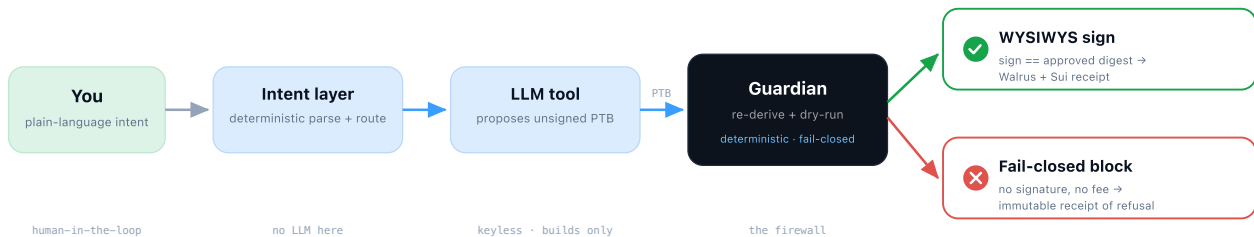


The agent builds, the Guardian verifies, you sign.

A plain-language intent compiles into one unsigned PTB. Before any signature, a deterministic Guardian re-derives the math and dry-runs the exact bundle — every value move runs the full gauntlet.

Intent → sign — the firewall pipeline

flow · 02



LLM proposes, deterministic Guardian re-derives and fail-closes — any single gate failing blocks the whole PTB.

The 14 Guardian gates — every PTB runs the gauntlet

gates · 03

GATE	WHAT IT RE-DERIVES — ONE LINE EACH
● value & integrity	
1 server caps	per-tx + per-day USD caps; invalid cap config blocks everything; borrow has its own inflow cap.
2 trusted price	real oracle value (SUI/USD live, USDC=\$1); no price → block. LST outflow uses a floor formula.
3 min-out re-derive	recompute swap min-output from on-chain decimals + same route; runs for every swap (anti-sandwich).
4 authoritative value	re-value from dry-run net deltas; block when outflow > 1.5× declared (outflow_mismatch).
5 WYSIWYS digest	approvedDigest = sha256(kindBytes); signed bytes must equal the reviewed kind.
● identity & correctness	
6 allowlist	every MoveCall {pkg::mod::fn} must be pre-approved; runs first so unknown targets never process.
7 action-shape	PTB call-set must match exactly one actionType template; one value action per PTB.
8 coin-type provenance	coinTypeIn/out verified on-chain via CoinMetadata; identity by TYPE, never display symbol.
9 injection provenance	per-field argProvenance; a derived recipient triggers a confirm gate (anti prompt-injection).
10 SuiNS lookalike	homoglyph-normalized edit-distance vs verified contacts (catches 888-l.sui-style spoofs).
11 staking constraints	LST coin-type vs curated map; provider-keyed shapes (afSUI ≠ haSUI cannot cross).
12 post-tx health factor	NAVI borrow/withdraw; devInspect HF < threshold (1.6) blocks; full-withdraw cannot leave debt.
● dependency fail-closed	
13 dry-run + effects	dry-run exact PTB bytes incl. gas; RPC/dry-run failure or unknown target blocks, never proceeds.
14 composite recipe	atomic single-sign for ANY allowlisted combo (send/swap/lend/stake + chaining): dynamic recipe · target multiset · recipient-aware anti-leak · delta/owner + dual caps.

One moat pattern, three places it shows up.

The LLM only ever **proposes**; deterministic code **verifies** and fail-closes. The same pattern guards trade preparation, multi-intent decomposition, and atomic composite recipes.

The moat — LLM proposes, deterministic code verifies

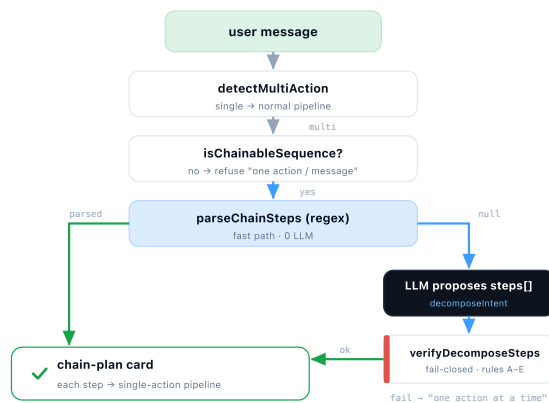
pattern · 04



Same router that guards a trade also guards a decomposition step and a composite leg — proposal and verification never share authority.

Hybrid multi-intent decompose

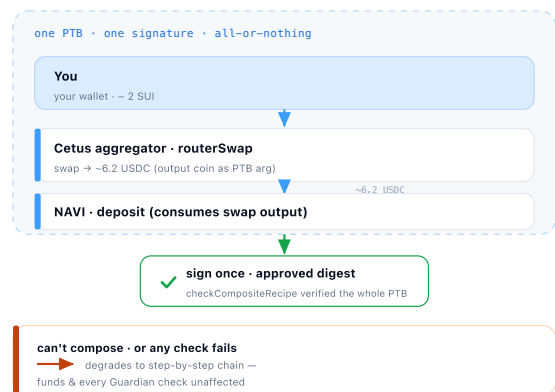
flow · 05



verifyDecomposeSteps rejects all on: **A** <2 steps · **B** category ∉ {swap,lend, stake, send} · **C** router disagrees · **D** step 0 not explicit amount · **E** a step hides a 2nd action.

Atomic composite — one signature

flow · 06



recipe swap_lend_v1: the swap-output coin feeds the deposit *inside* the PTB — a Move abort in either leg reverts both.